

AI Workspace

Agent Suite

Google Workspace MCP · LangGraph ReAct · Gmail + Calendar APIs

- Refund Email Agent — automated triage & reply
- Calendar Agent — intelligent scheduling & RSVP
- ReAct reasoning loop via LangGraph StateGraph
- stdio-based MCP transport, no external middleware

Key System Requirements

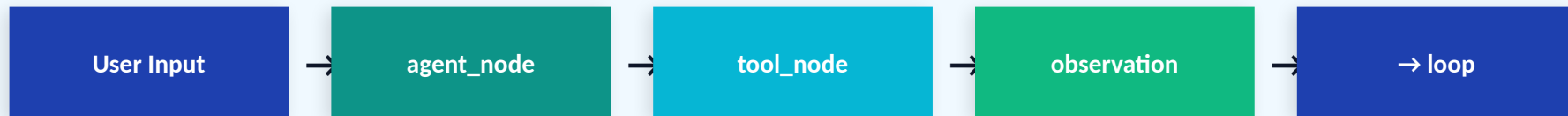
Software Requirements

- Python 3.11+
- OpenAI GPT-4o API access
- LangGraph
- langchain-openai
- langchain-core
- langchain-mcp-adapters
- workspace-mcp package

Google Cloud Requirements

- Google Cloud Project
- Gmail API enabled
- Google Calendar API enabled
- OAuth 2.0 Desktop App credentials

Core Architecture Requirements



Until task is complete

Key Components

- LangGraph StateGraph
- ToolNode
- OpenAI GPT-4o
- MCP Server
- workspace-cli subprocess tools

MCP Transport

- stdio-based local subprocess
- JSON-RPC over stdin/stdout
- No external middleware required

Installation Steps

01

Clone MCP Repository

```
git clone https://github.com/taylorwilsdon/google_workspace_mcp  
cd google_workspace_mcp
```

02

Install CLI + MCP

```
uv tool install .  
pip install workspace-mcp
```

03

Install Python Dependencies

```
pip install langgraph langchain-openai langchain-mcp-adapters
```

Google Cloud Console Setup

1

Create Google Cloud Project

- Open:
console.cloud.google.com
- Create a new project

2

Enable APIs

- Gmail API
- Google Calendar API

3

Configure OAuth Consent Screen

- Choose: External
- Add your Google account as a test user

Tip: Save your project ID — you'll need it for API scopes and billing setup.

OAuth Scopes Required

Gmail

gmail.modify

Read & modify emails

gmail.send

Send threaded replies

Calendar

calendar

Read calendar data

calendar.events

Create / update / delete events

Purpose

Read emails · Send replies · Read calendar events · Create / update / delete events

How to Obtain OAuth Credentials

Create OAuth Credentials

Inside Google Cloud Console:

- 1 APIs & Services
- 2 Credentials
- 3 Create Credentials
- 4 OAuth Client ID
- 5 Application Type: Desktop App

Keys Provided by Google

Client ID

Public identifier for your app

Client Secret

Private key — keep secure

These are required by workspace-mcp

Export Environment Variables

```
export OPENAI_API_KEY=sk-...
```

```
export GOOGLE_OAUTH_CLIENT_ID=<your-client-id>
```

```
export GOOGLE_OAUTH_CLIENT_SECRET=<your-client-secret>
```

```
export OAUTHLIB_INSECURE_TRANSPORT=1
```

 OAUTHLIB_INSECURE_TRANSPORT=1 is for local development only

 Never hardcode secrets in source code — use environment variables or a secrets manager

Verify Installation

Verify CLI Installation

```
$ workspace-cli list
```

→ Lists all available MCP tools

Verify OAuth Connection

```
$ workspace-cli call list_calendars
```

→ Tests OAuth authentication · Calendar API access · MCP server connectivity

Running the Agents

Refund Email Agent

```
$ python refund_agent.py
```

Functions:

- Search refund emails
- Read email content
- Classify requests
- Generate replies
- Send threaded responses

Calendar Agent

```
$ python calendar_agent.py
```

Functions:

- Read calendar events
- Create / update / delete meetings
- Find free time slots
- RSVP handling

Key Learning Concepts

ReAct Pattern

LangGraph StateGraph

MCP Protocol

Tool Binding

OAuth 2.0

stdio Transport

ToolNode Loop

Main Educational Goal

Build production-style AI agents that reason, call tools, interact with real APIs, and maintain multi-turn memory.